

Advantages and Disadvantages of Lithium-Ion Batteries

Advantages	Disadvantages
• Electrodes of a lithium-ion battery are made of lightweight lithium and carbon; hence, these are generally much lighter than other types of rechargeable batteries of the same size. • Lithium, being a highly reactive element, will store a lot of energy in its atomic bonds, leading to very high energy density lithium-ion batteries. A typical lithium-ion battery can store 150 watt-hours of electricity in one kilogram of battery. • Battery pack can store, perhaps, 100 watt-hours per kilogram, and a lead-acid battery can store only 25 watt-hours per kilogram. • These hold their charge for long as a lithium-ion battery pack loses only about five per cent of its charge per month, compared to a twenty per cent loss per month for NiMH batteries. • In comparison to other batteries, lithium batteries show memory effect, that is these do not completely discharge before recharging within a required time. • Lithium-ion batteries can sustain hundreds of charge/discharge cycles. • Lithium-ion is a low maintenance battery.	• These batteries start degrading as soon as they leave the factory, and will only last two or three years from the date of manufacturing whether used or not. • These are extremely sensitive to high temperature, and heat can cause lithium-ion battery packs to degrade much faster than they would normally do. • A lithium-ion battery becomes useless if it is left to be completely discharged. • A lithium-ion battery pack must have an onboard computer to manage the battery; however, this makes the batteries even more expensive than they already are. • In case a lithium-ion battery pack fails, there is a small chance that it may burst into flame.

high voltage making the cells produce 3.7 volts, which is much higher than the 1.5 volts typical of a normal AA alkaline cell.

Lithium-ion charge/discharge basics

Charging and discharging characteristics of lithium-ion battery are important, especially for its safe operation and long-term performance. To manage smooth and safe charging and discharging processes of lithium-ion battery, a typical battery management chip is incorporated into the battery pack. This way the user can plug the battery into a charger and leave it to charge with the knowledge that it does not have to be unplugged after a certain time. Moreover, the battery management chip also ensures that the battery is not discharged beyond limits so that it can be recharged easily. The focus is to ensure that the battery management pack understands the exact state of battery charge.

Charging lithium-ion batteries is different from charging Ni-Cads or NiMH batteries. Charging of lithium-ion batteries is voltage-sensitive rather than current-sensitive. Charging of lithium-ion batteries is same as charging of lead-acid batteries, only difference being lithium-ion batteries have a higher voltage per cell. These also require much tighter voltage tolerance on detecting full charge, and once fully charged these do not allow or require to be trickle or float charged.

Consumer-oriented lithium-ion batteries mostly charge to a voltage of 4.2 volts per cell with a tolerance of around $\pm 50\text{mV}$ per cell. Charging of batteries beyond this voltage

causes stress to the cell, resulting in oxidation, which reduces service life and capacity of the battery. Therefore it is important to be able to detect the full charge state of battery accurately because lithium-ion batteries do not tolerate overcharging, which can also cause safety issues.

For lithium-ion batteries, the amount of charge retained by the battery or cell against the amount of charge entering the cell (known as charge efficiency) is high, and charge efficiencies of around 95 to 99 per cent can be achieved with low-level temperature rise of cells. Charging process of lithium-ion batteries can be separated into two main stages, as given below.

Constant current charge. In the first stage of charging a lithium-ion battery or cell, charge current is controlled, which typically is between 0.5C and 1.0C. Capacity of the battery is commonly rated at 1C, meaning that a fully charged battery rated at 1Ah should provide 1A for one hour, and for a 2000mAh battery, charge rate would be 2000mA for a charge rate of 1C.

For consumer-based lithium-cobalt (LCO) cells and batteries, charge rate of a maximum of 0.8C is recommended. Thus, during charging process, voltage across the lithium-ion cell increases for constant current charge, and charge time may be around an hour.

Saturation charge. After some time, voltage peaks at 4.2 volts for an LCO cell, and at this stage the cell or battery must enter a second level of charging, known as saturation charge. Constant voltage of 4.2 volts is maintained and current steadily falls. End

of the charge cycle is reached when current falls to around ten per cent of rated current. In this case, charge time may be around two hours depending on cell type.

Lithium-ion batteries' charging/discharging characteristics show improvements over its competitors, and have been shown to be able to withstand over one thousand charge/discharge cycles but still be able to hold eighty per cent of initial capacity.

NiCd batteries offer up to around 500 cycles, which is very much dependent on the way batteries are used, and a badly treated cell may only give fifty or hundred.

NiMH cells are even worse, and this is one of the main areas receiving development. These are only able to give 500 cycles at the very best before their capacity drops to eighty per cent of initial charge rating.

Due to a number of significant advantages associated with lithium-ion batteries in comparison to competing technologies, these batteries are becoming popular but are not without limitations. The advantages and disadvantages of lithium-ion batteries are summarised in Table 1.

Lithium-ion battery charging precautions

In view of the amount of energy stored in lithium-ion batteries and the nature of their chemistry, etc, it is necessary to ensure that the batteries are charged in appropriate manner and with an appropriate charger and equipment. Battery chargers or battery packs have inbuilt mechanisms to prevent damage and dangers due to overheating. Mechanisms required for charging